

Model Optimization, Evaluation, and Workflow

Isaac Mehlhaff

PLSC/DATA 40501, Fall 2026

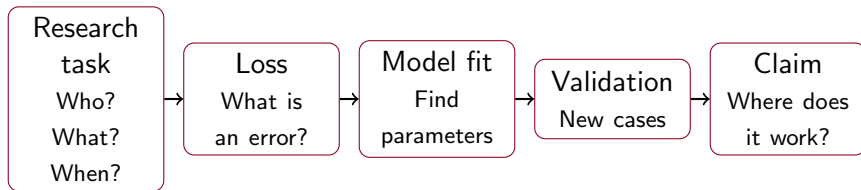
Outline

1. Task and Baselines
2. Risk, Loss, and Evaluation
3. Optimization
4. Complexity and Tuning
5. Evaluating Predictions

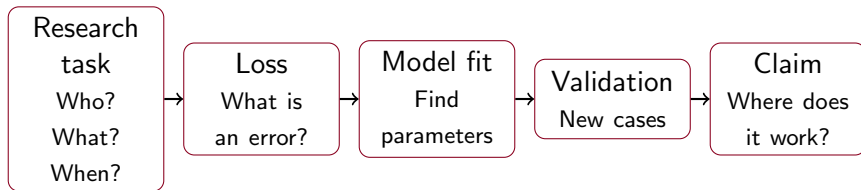
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Full Workflow



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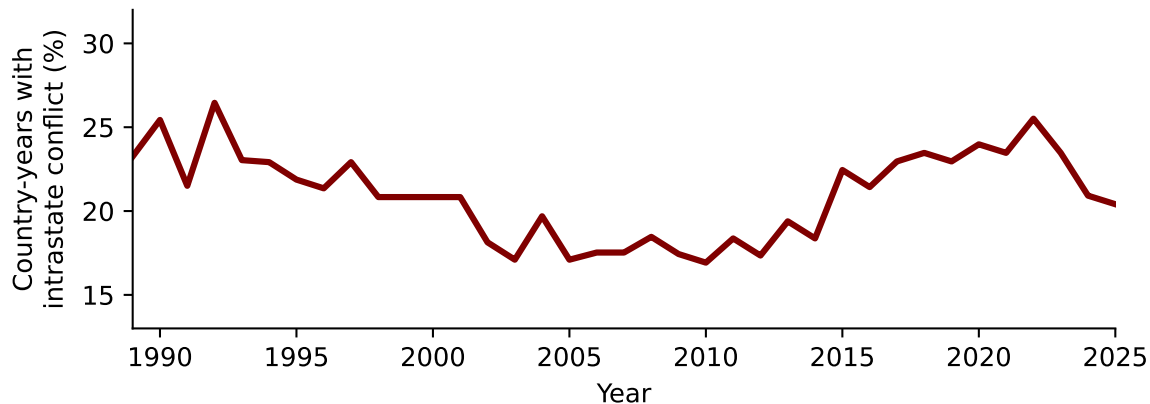
Optimization: which parameters minimize the training objective?

$$\hat{\theta} = \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(\mathbf{x}_i))$$

Evaluation: how does the whole procedure do on new cases?

$$\hat{R}_{\text{new}} = \frac{1}{m} \sum_{j=1}^m \ell(y_j^{\text{new}}, \hat{f}(\mathbf{x}_j^{\text{new}}))$$

Running Example: Forecasting Intrastate Conflict



Uppsala Conflict Data Program, by country-year

Establishing Baselines

Many reasonable baselines, depends on task

- Majority class, mean, median, lagged value, standard parametric model, human/institutional procedure being replaced, ablations of the final model

```
observed = development["sb_intrastate_exist"]
persistence = development["prior_conflict"]
onset = (persistence == 0) & (observed == 1) # peace last year, conflict this year

print(round(observed.mean(), 3), # % country-years with conflict
      round((observed == persistence).mean(), 3), # accuracy of repeating previous year
      round((persistence[onset] == 1).mean(), 3), # % conflict onsets flagged
      sep = "\n")

## 0.204
## 0.935
## 0.0
```

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Risk and Loss Review

$\ell(y, \hat{y})$ assigns a cost to predicting $\hat{y}$ when the truth is y

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Population risk of *any* candidate rule f :

$$R(f) = \mathbb{E}_{p^*(X,Y)} [\ell(Y, f(X))]$$

Minimized by f^* , which we cannot compute
because $p^*(X, Y)$ is unknown

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Empirical risk of the same f on data $\mathcal{D}$:

$$\hat{R}(f, \mathcal{D}) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(\mathbf{x}_i))$$

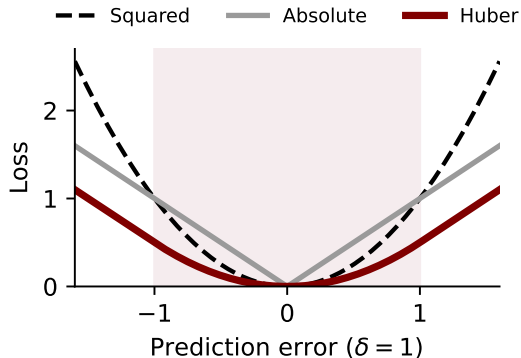
Minimized over a function class $\mathcal{H}$ by $\hat{f}_{\mathcal{D}}$

Basic Regression Losses

With $e_i = y_i - \hat{y}_i$ fatalities in year t :

- RMSE = $\sqrt{\frac{1}{n} \sum_i e_i^2}$ magnifies large misses
- MAE = $\frac{1}{n} \sum_i |e_i|$ weights misses equally
- Huber loss:

$$h_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta \\ \delta(|e| - \frac{\delta}{2}), & |e| > \delta \end{cases}$$



Two Sources of Excess Risk

f^* is best possible rule, $f_{\mathcal{H}}^*$ is best rule in class $\mathcal{H}$, $\hat{f}_{\mathcal{D}}$ is rule fitted to sample $\mathcal{D}$:

$$R(\hat{f}_{\mathcal{D}}) - R(f^*) = \underbrace{R(f_{\mathcal{H}}^*) - R(f^*)}_{\text{approximation error}} + \underbrace{R(\hat{f}_{\mathcal{D}}) - R(f_{\mathcal{H}}^*)}_{\text{estimation error}}$$

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- **Approximation:** shallow tree is too rigid for how prior violence relates to future conflict
→ enrich $\mathcal{H}$ (shrinks first term but inflates second term)

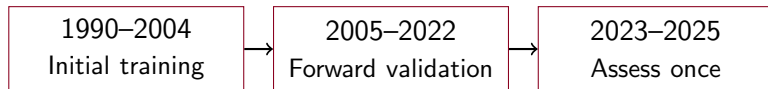
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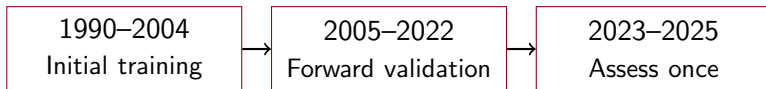
- **Approximation:** shallow tree is too rigid for how prior violence relates to future conflict
→ enrich $\mathcal{H}$ (shrinks first term but inflates second term)
- **Estimation:** deep tree fits to country-year noise → regularize or enrich $\mathcal{D}$

Sample Splitting

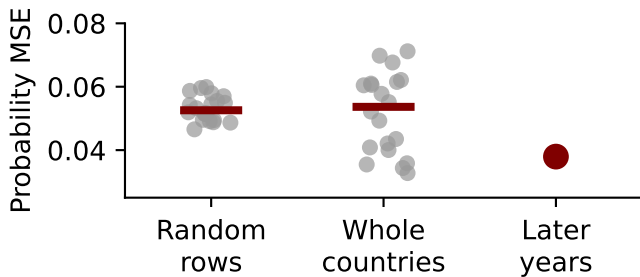


- **Training** estimates parameters
- **Validation** selects model classes, hyperparameters, thresholds, stopping points
- **Test** assesses once every choice is made

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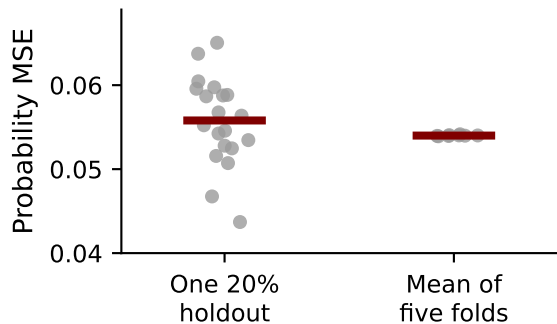


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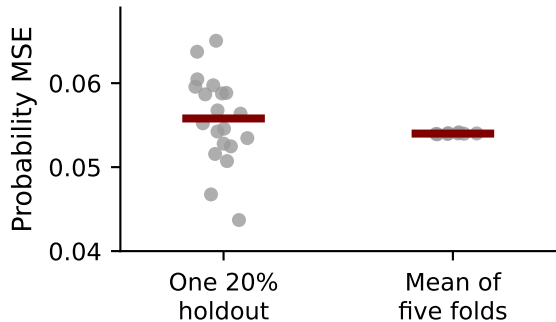
Cross-Validation

Test sets limit training data, give noisy estimates. CV removes sensitivity to partition

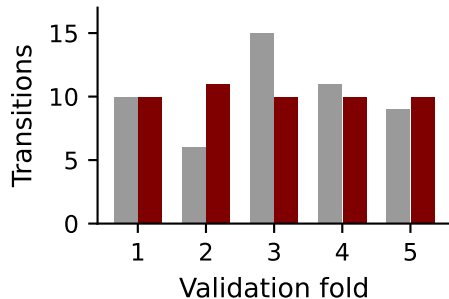


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Stratify folds for rare events; gray folds are random, red stratified



Variants of CV

Design	Cost/limitation	K	Train per fit	Fits
Repeated K -fold	More model fits	2	50%	2
Leave-one-out	N fits, overlapping folds	5	80%	5
Grouped folds	Fewer independent groups	10	90%	10
Rolling, expanding windows	Early fits have less history	N	$(N - 1)/N$	N

- Larger K fits on more cases but costs more, LOO rarely beats moderate K
- Repeat partitions for sensitivity
- Break down by country-year to identify sources of error

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Objectives and Optimization

Logistic objective $J(\beta_0, \beta)$, with $z_i = \beta_0 + \mathbf{x}_i^\top \beta$ and $p_i = (1 + e^{-z_i})^{-1}$:

$$J(\beta_0, \beta) = \frac{1}{n} \sum_{i=1}^n \underbrace{[\log(1 + e^{z_i}) - y_i z_i]}_{\text{neg. Bernoulli log-lik.}}$$

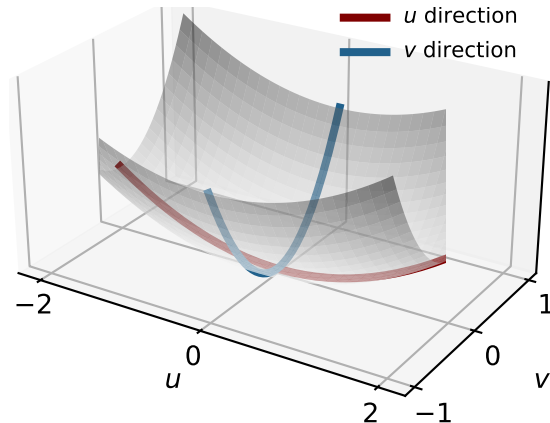
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- Fitting returns $\hat{\beta} = \operatorname{argmin}_{\beta} J(\beta)$, equivalently the maximum likelihood estimate
- **Stop** on gradient norm $\|\nabla J_k\| < \epsilon$ (i.e. gradient is sufficiently close to flat), parameter change, or a specified budget

Gradient and Hessian



$$J(u, v) = \frac{1}{2}(u^2 + 8v^2)$$

$$\nabla J = (u, 8v)^\top$$

- Vector, first partial derivatives
- Slopes

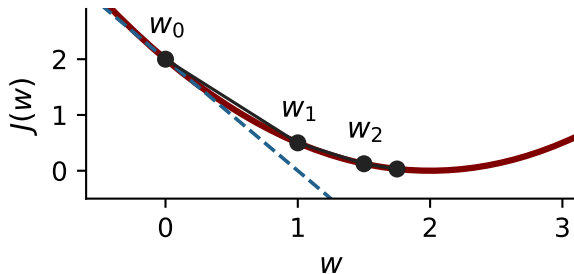
$$H = \text{diag}(1, 8)$$

- Matrix, second partial derivatives
- Curvatures

Gradient Descent

1. Start at a guess θ_0

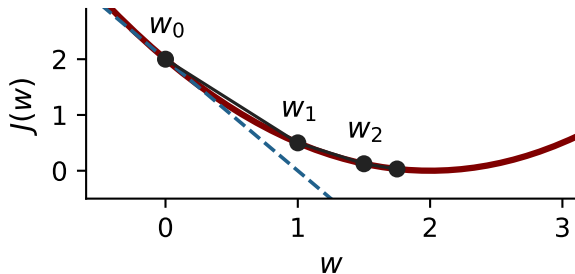
$$J(w) = \frac{1}{2}(w - 2)^2, \eta = 0.5$$



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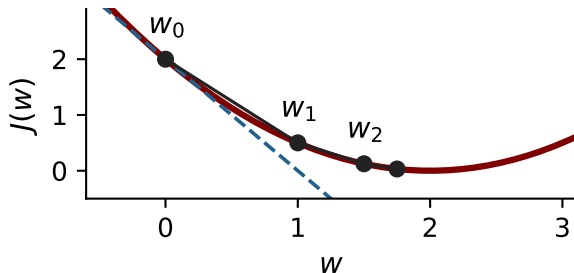


Gradient Descent

1. Start at a guess θ_0
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3. Step other way, scaled by **learning rate** η :

$$\theta_{k+1} = \theta_k - \eta \nabla J(\theta_k)$$

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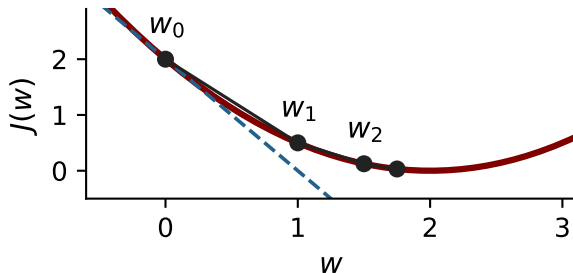
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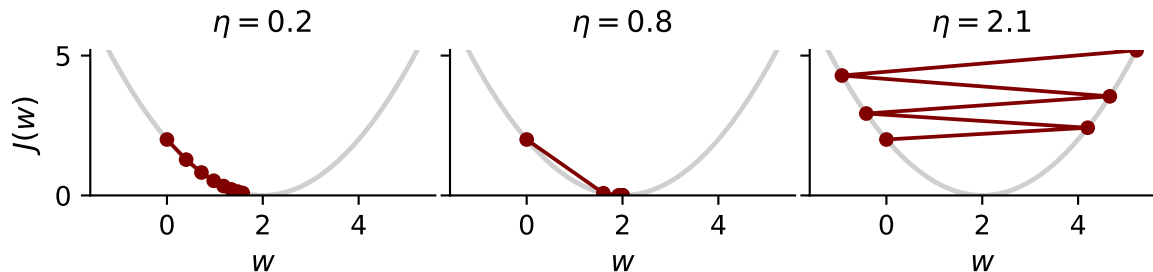
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4. Repeat until $\|\nabla J\| \leq \epsilon$, or budget runs out

$$J(w) = \frac{1}{2}(w - 2)^2, \eta = 0.5$$



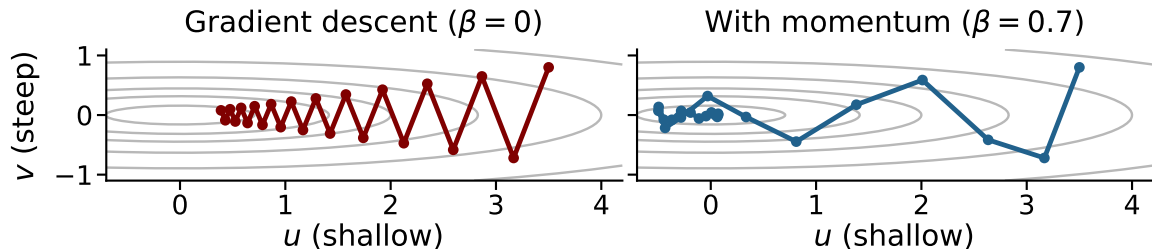
Step Size



Too small wastes iterations, too large climbs back out

Learning rate schedule to slow down descent, momentum to accelerate in consistent direction

Momentum



Much steeper in v , so a good step in v is very slow in u

Quadratic objective produces “zig-zag” pattern

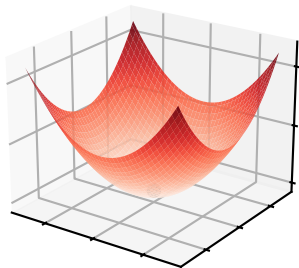
Momentum keeps part of the last step, so oscillations cancel, faster convergence:

$$m_{k+1} = \beta m_k + \nabla J(\theta_k), \quad \theta_{k+1} = \theta_k - \eta m_{k+1}$$

Zero Gradient $\neq$ Minimum

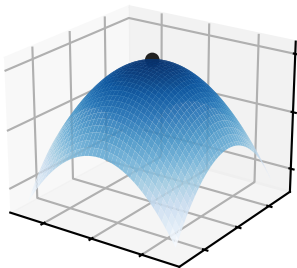
Minimum

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



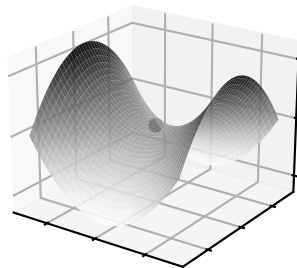
Maximum

$$H = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$



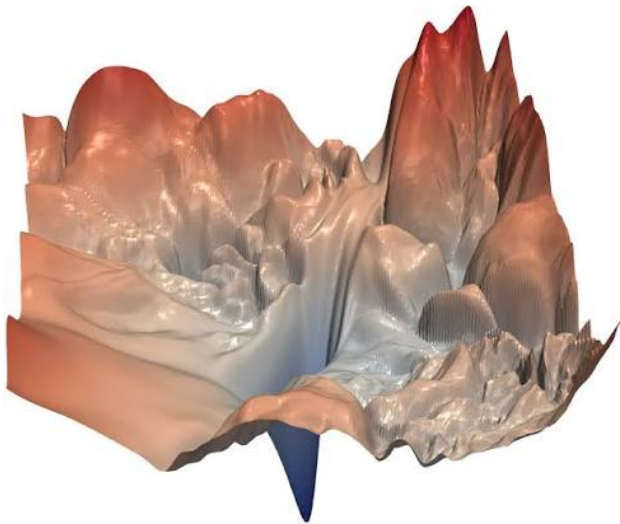
Saddle

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Neural Network Loss Topology

ResNet-56: CNN for small
images (Li et al. 2018)

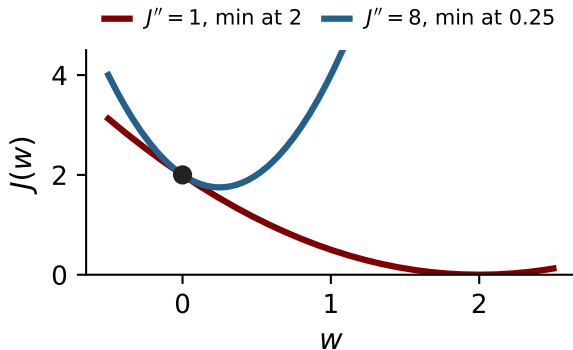


Incorporate Curvature

Same slope at $w = 0$, minima far apart

Newton's method: fit a quadratic at θ_k ,
step toward its minimum:

$$\theta_{k+1} = \theta_k - \eta_k H_k^{-1} \nabla J(\theta_k)$$



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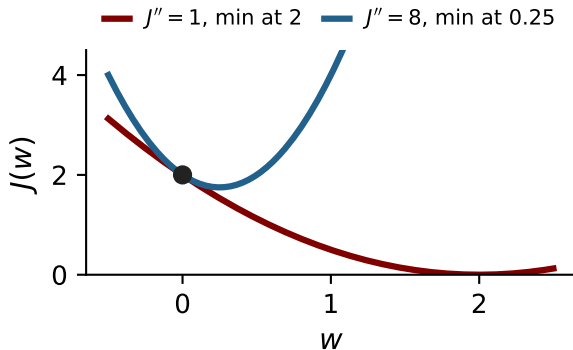
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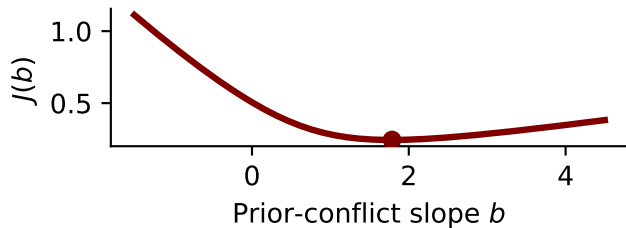
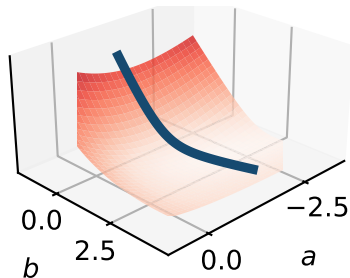
H_k^{-1} undoes any skew in local curvature, so $\eta_k = 1$ hits the min exactly

Quadratic approximation is exact for OLS, so one Newton step hits solution

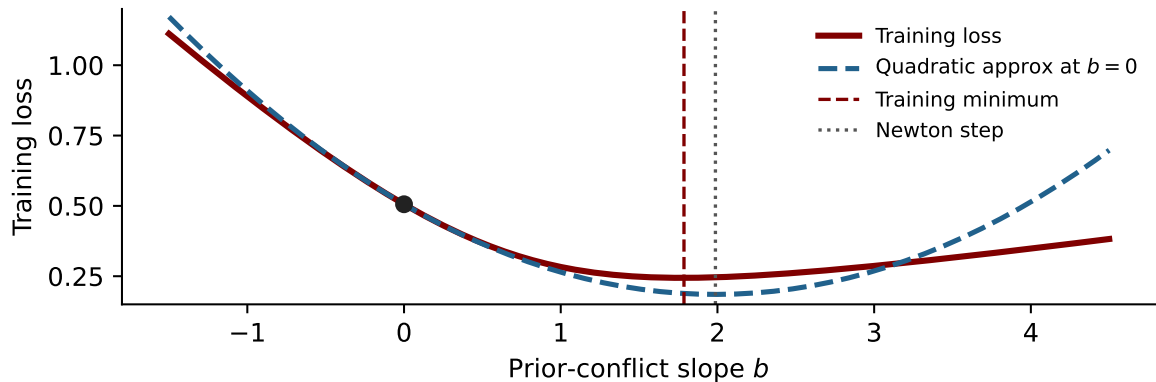


Conflict Model Loss

Intercept a and standardized prior-conflict slope b . Set $a = \log(\bar{y}/(1 - \bar{y}))$ so only b varies



One Newton Step



Overshoots slightly because true loss is not quadratic, so iterate until convergence

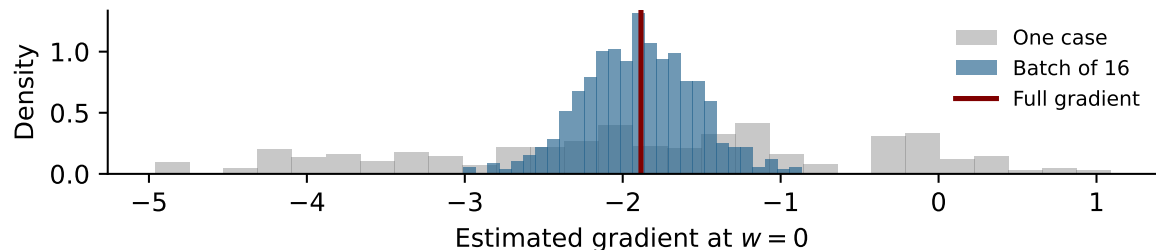
Stochastic Gradient Descent

So far, we have worked with **full-gradient** steps

SGD: randomly select one case or mini-batch, compute only that gradient, iterate

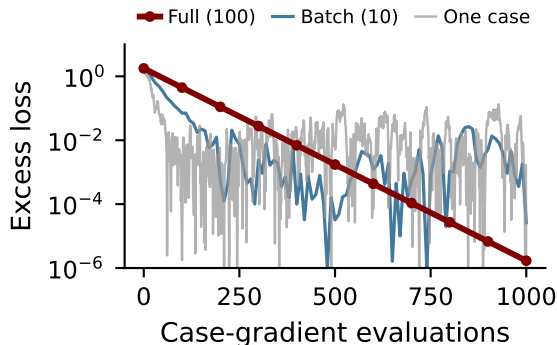
- Noisier but much faster

Epoch: one full pass through the data



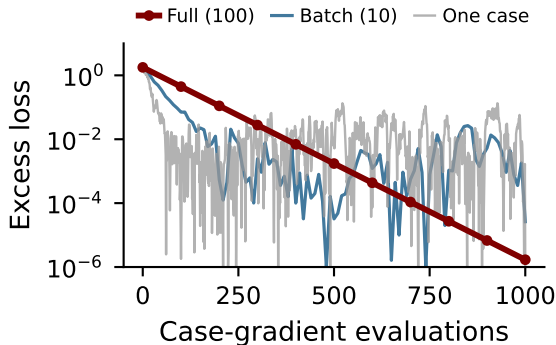
Batching and Scheduling

Small batches improve fast, then fluctuate.
Progressive batching can manage this

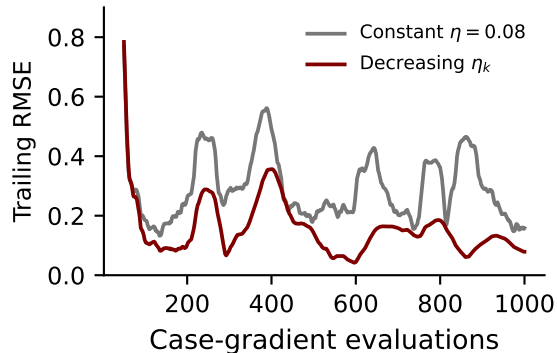


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Constant η more sensitive to noise, scheduled η (or backtracking) more robust



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Bias-Variance Tradeoff

At fixed $\mathbf{x}$, let $\theta = E[y \mid \mathbf{x}]$ and $\hat{\theta}$ the fitted forecast, with expectations over training samples:

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

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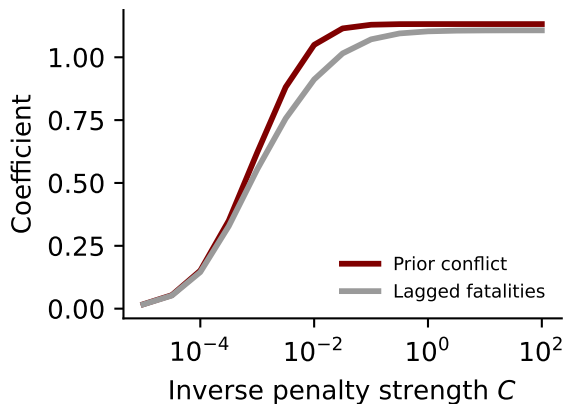
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We can accept greater bias as long as variance falls by more $\rightarrow$ logic of regularization

Regularization

$$\hat{\beta}_\lambda = \operatorname{argmin}_{\beta} \left\{ \hat{R}(\beta, \mathcal{D}) + \lambda \|\beta\|_2^2 \right\}$$

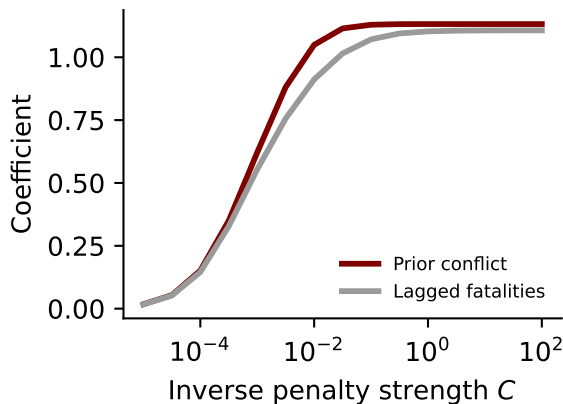
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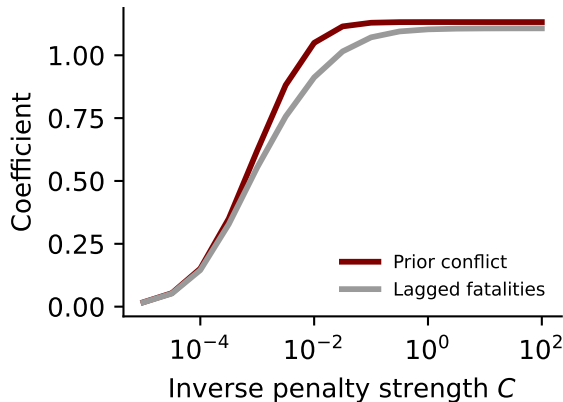
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- Larger λ shrinks slopes (in scikit-learn, smaller C implies larger λ)
- **Early stopping** also regularizes implicitly
- Penalty punishes small-unit features
→ standardize data



Hyperparameter Tuning

Fitting cannot choose λ

- Would always choose $\lambda = 0$ because regularization makes training loss worse by construction
- λ is a **hyperparameter**, tuned on the validation set

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Model must be trained using each candidate, so search is a **budgeting problem**

- **Grid** evaluates pre-specified combinations, cost exponential in number of hyperparameters
- **Random** tries more values, usually beats grid when few hyperparameters matter
- **Sequential or Bayesian** models loss across settings, then probes promising regions
- **Successive halving** distributes a small budget and focuses on the best half

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tn, fp, fn, tp = confusion_matrix(
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	No flag	Flag
No conflict	TN = 585	FP = 11
Conflict	FN = 25	TP = 163

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Recall R (TPR): $\frac{TP}{TP+FN} = 86.7\%$

Precision P : $\frac{TP}{TP+FP} = 93.7\%$

Accuracy: $\frac{TP+TN}{n} = 95.4\%$

F_1 : $\frac{2 \cdot P \cdot R}{P+R} = 90.1\%$

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Default is $\tau = 0.5$. More principled methods: 1) analytical, 2) Youden's J , 3) equal error rate

Choosing Actions Based on Costs

Choose action a to minimize expected loss $\rho(a \mid \mathbf{x}) = \sum_y \ell(y, a) P(y \mid \mathbf{x})$

$\ell(y, a)$	Do not flag	Flag
No conflict	0	1
Conflict	4	0

False alarm costs $c_{FP} = 1$,
missed conflict costs $c_{FN} = 4$

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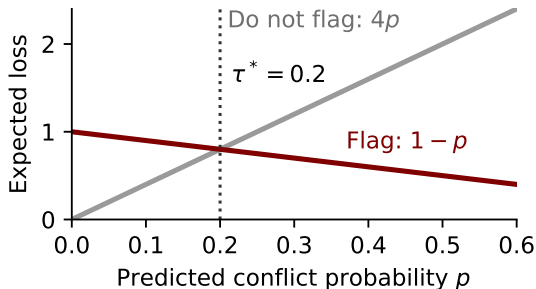
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False alarm costs $c_{FP} = 1$,
missed conflict costs $c_{FN} = 4$

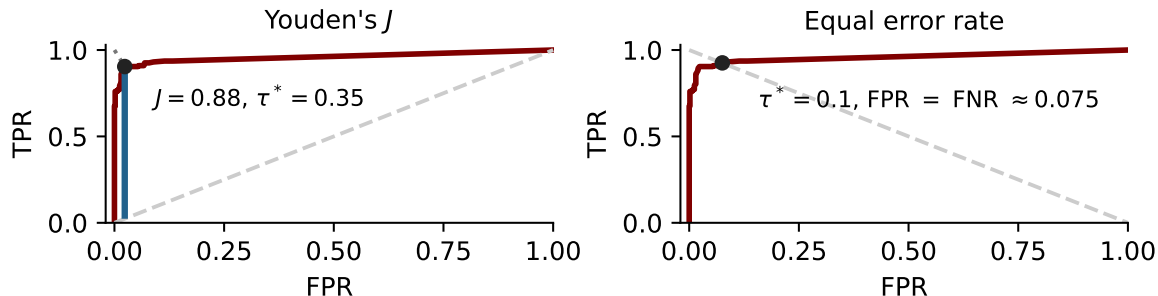
For prediction p , flagging costs $1 - p$, not
flagging $4p$, so flag when $1 - p < 4p$:

$$\tau^* = \frac{1}{1 + c} = \frac{1}{5} = 0.2$$



ROC Curves and Selecting τ^*

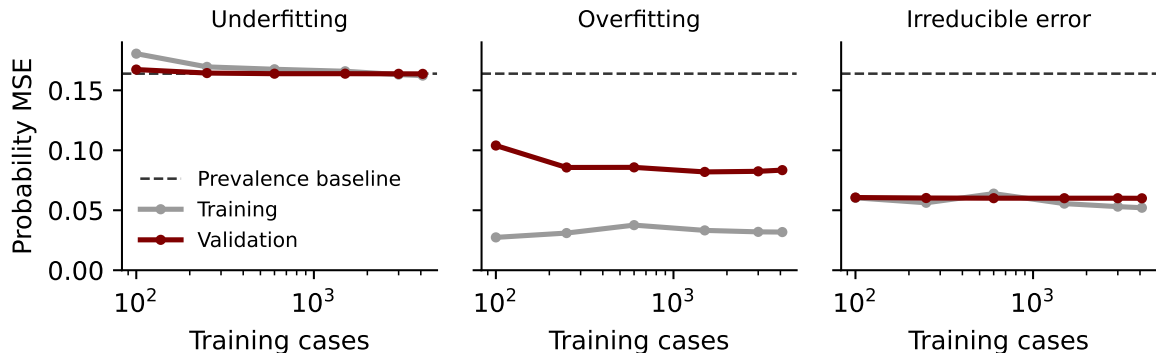
ROC curve plots TPR against FPR for all τ . AUC insensitive to class prevalence



Youden's J maximizes $\text{TPR} - \text{FPR}$ (blue gap above random-chance diagonal)

Equal error rate is the intersection with $\text{TPR} = 1 - \text{FPR}$

Diagnosing Learning Curves



Underfitting: nothing learned $\rightarrow$ enrich $\mathcal{H}$

Overfitting: gap still closing $\rightarrow$ regularization or more data

Irreducible error: far below baseline and flat $\rightarrow$ more/better features, better labels

Data/Prediction Misalignment

Leakage: information from outside training data enters fitting or evaluation

- Year- t fatalities or later revisions enter a forecast made at $t - 1$
- Imputation or scaling learns from validation and test rows
- Predictors screened against all outcomes before folds exist
- Same politician has speeches in training and test, when the claim concerns new politicians

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Distribution shift: evaluation cases stop resembling cases where the forecast gets used

- Conflict becomes more common after a regional shock
- Peace agreement changes how prior violence predicts next year

Validation relies on evaluated cases $\approx$ predicted cases

- Complicated by institutions and strategic actors